The Curated Database: from the v2 Function-Keyed Store to the v3 True-Friend Store

local_mixing project note

September 28 – October 2, 2026

Summary of the curated-DB redesign: old vs. new structure and operation, the definitional variants, the experimental record, and the rationale for continuing with modes (c) and (b′). Companion artifacts: `CURATED_DB_FINDINGS.md` *(pre-redesign diagnosis),* `runs/ncdb_ab_20261001/RESULTS.md` *(experiment tables),* `runs/ncdb_random_20261001/RESULTS.md` *(ablation runs).*

1 Setting

Stage 3 (`db_mixing`) replaces sampled windows of the circuit with equivalent spellings drawn from two frozen stores: the **regular** store (`frozen_m1_m11`, the complete g57 replacement corpus through 11 gates, function-keyed) and the **curated** store, whose job is to offer *meaningfully different* alternatives for a window—above all, equal-length re-spellings, the staple of in-place mixing. The redesign concerns only the curated store; the regular store and the move loop are unchanged.

2 The old store (v2, function-keyed)

Construction (`build_curated_full from-frozen-identities-v2`):

1. For every multi-member class of the regular store (9.68M classes: functions with ≥ 2 stored spellings a, b of ≤ 11 gates), form the pair identity $a \cdot \mathrm{rev}(b)$; canonicalize; cancel adjacent duplicates.
2. Deduplicate identities by dihedral orbit: 20.0M words $\to$ 7.47M orbits.
3. For each orbit word of length n: enumerate every direction $\times$ rotation $\times$ split. Each split $P \,|\, S$ emits *both* P and $\mathrm{rev}(S)$ under the *function key of the prefix* (canonical-polynomial hash), rewired into the prefix's canonical wire frame and gate-canonicalized.
4. Keep one lexicographic representative per (identity, key, gate count)—the "sibling collapse"—then byte-exact global dedup.

Shape. 15.55M keys / 276M candidates. Structural facts established during the diagnosis: every pair identity simplifies to ≤ 12 gates (multi-member classes concentrate at ≤ 6-gate members; longer pairings cancel), hence candidates cap at 11 gates; **keys exist only for functions of minimal length** ≤ 6; long candidates pile up under near-trivial keys (the single-gate key held 7.7M spellings, 245 MB; the identity-function key 2.6M).

Operation. The curated-exhaust cascade probes the curated store first, at every window length, by the window's *function* key (forward direction); any stored spelling of that function is a candidate; the regular store is consulted only on a complete curated miss.

Why it failed.

F1. **The sibling collapse was wrong at the midpoint**: there P and $\mathrm{rev}(S)$ are the two genuinely different halves (a and b), and one was dropped. 48% of keys held a *single* candidate—usually the window itself—although a different same-length spelling provably existed for 95% of them.

F2. **No difference guarantee**: pools mixed rotation debris, commuting re-spellings, and the window itself; once the canonical-order identity guard landed, 43% of moves were re-spelling rejections.

F3. **Catastrophic lookup cost**: mean candidates per splice choice 327k–660k; throughput fell from ∼100k moves/min to ∼100k moves per 30–60 min.

F4. **No provenance** for why a candidate suited a window.

3 The new notion: true friends

With involutive gates, $C^* = C^{-1} = \mathrm{rev}(C)$.

Strict definition. D is a *true friend* of C (spellings of one function f, $\delta = \mathrm{minlen}(f)$) when the closed loop $C \cdot D^*$ is *locally minimal except for a sanctioned chord* between its junctions ID and P_f: for every pair of prefix-permutations u, v on the loop, the true Cayley distance equals the arithmetic minimum over the four sanctioned routes (two arcs along the loop; two routes through a δ-gate geodesic chord). The per-circuit half of this—every window of C sits exactly on its sanctioned routes—is **chord-tightness**; for a geodesic spelling it reduces to classical local geodesicity, and for a non-minimal spelling it allows exactly the slack the chord explains. By construction: the two prefix paths never re-meet away from the junctions, so self-pairs and all commuting reorders are excluded (F2 dies definitionally); the identity-function key cannot exist; provenance is the definition itself (F4).

Relativized routes. When the *key* circuit is not required to be tight (below), junction routes are priced with proven distances of its actual prefixes rather than raw gate counts—the original "shortest path through ID or P_C" phrasing; on tight circuits this coincides with the arithmetic form.

Leeway. A graded relaxation lets a check miss its sanctioned minimum by at most t gates, with $t = 2$ for pair total length ≥ 9, $t = 1$ for 7–8, else 0.

The oracle. Distances are decided by a layered oracle: exhaustive exact tables for minlen ≤ 3 (all 1/2/3-gate words up to relabelling—$1 + 31 + 1{,}639$ functions, cross-validated against the pool census) plus regular-store members as shortcut witnesses. Checks with threshold ≤ 4 are exact; above that, a pass means *no shortcut witnessed*—the same evidence the replacement-db adversary holds. The layering was forced by a discovery: the regular store is a replacement table, *not* a minimality oracle (the single-gate function's class stores only 5- and 6-gate re-spellings).

4 The new store (v3 "ncdb", circuit-keyed)

Pools. The v2 enumeration re-ran *without* the sibling collapse: same 7.47M identities, all exact-distinct arcs kept—779.2M candidates over the same 15.55M function keys, **zero single-candidate pools** (F1 fixed at the source).

Pairs. One pass over the pools (reservoir ≤ 48 members per pool $\times$ length, $\leq 2{,}500$ pairs per pool; pools ≤ 64 members—96% of keys—are exhaustive) evaluates each sampled pair under the layered oracle and emits every qualifying *ordered* pair $(C \to D)$ into per-mode stores:

mode	key C	friend D	cross	keys	pairs
a	tight	tight	exact	0.44M	0.83M
c	**unconstrained**	**tight**	**exact**	**1.96M**	**3.30M**
v2	unconstrained	slack $\leq t$, small exact	exact	2.54M	5.12M
c1	unconstrained	as b′, leeway ≤ 1	≤ 1	4.33M	9.42M
v1	unconstrained	tight	$\leq t$	12.05M	28.0M
b′	**unconstrained**	**slack $\leq t$, win. ≤ 4 exact**	$\leq t$	**16.87M**	**63.1M**
b	unconstrained	slack $\leq t$	$\leq t$	18.73M	194.2M

Containments: a $\subseteq$ c $\subseteq \{$v1, v2, c1$\} \subseteq$ b′ $\subseteq$ b. The *unconstrained key* is the deliberate asymmetry: a junky window should be replaceable by a certified friend; the relation is therefore directed, and splices drift toward tight material.

Records and keys. A record is $\mathrm{xxh3}_{128}$(canonical circuit blob of C) $\|$ blob of D: the key is the *circuit*, not the function—different spellings of one function are different keys, each owning only its certified friends, stored in the same canonical wire frame the function key would induce (the decode path is unchanged). Stores are direction-complete, with no min-direction fold. Friend lists are small by construction: (c) median 1 friend/key (mean 1.68), (b′) median 2 (mean 3.74).

Runtime. `FROZEN_CURATED_KEY=circuit` switches the curated stage (and the leakage-repair QC path) to the circuit key, computed byte-identically to the builder, with a round-trip-verified g57 encoder so non-g57 windows fail closed to a miss. Decode, selection, guards and splice are untouched; misses are the common case and cost a Bloom probe ($\sim$0.39% FP).

5 The experimental record

- **Probe.** Pool material is $\sim$85% tight at the short/geodesic end but 2.5–3.5% at 7 gates and $\sim$0% at 9–11 gates—most of the 779M-candidate mass is dead weight under any mode. Among tight pairs, strict passes are rare over hard functions ($\sim$0.8% at $(6,6)$, $\delta = 6$) and plentiful over easy ones ($\sim$80% at $(5,5)$, $\delta \leq 2$). The b-vs-b′ gap at $\geq$ 6-gate friends is almost entirely friends carrying compressible $\leq$ 4-gate windows—the hazard the b′ floor removes.
- **Full-profile A/B** (mini-sandwich TDP, 3.65M moves, same seed/binary): both arms finished in **27 min** (v2-store runs needed $>$ 1 day per 1M moves). Mean candidates per choice 1.5–2.2 (vs 327k); identity-skips at matched 700k moves 232k/224k vs 311k (old baseline) and 593k (old fixed); curated splices are two-thirds *equal-length* (22k 4$\leftrightarrow$4 + 27k 5$\leftrightarrow$5 for (c); 163k 5$\leftrightarrow$5 for (b′))—the move class v2 could not honestly serve.

- **Ridge vs the old store** (stage-2 rows vs 1M-move states, identical grids): old 0.3409; (c) 0.3267/0.3334 (two seeds); (b′) 0.3348/0.3318.

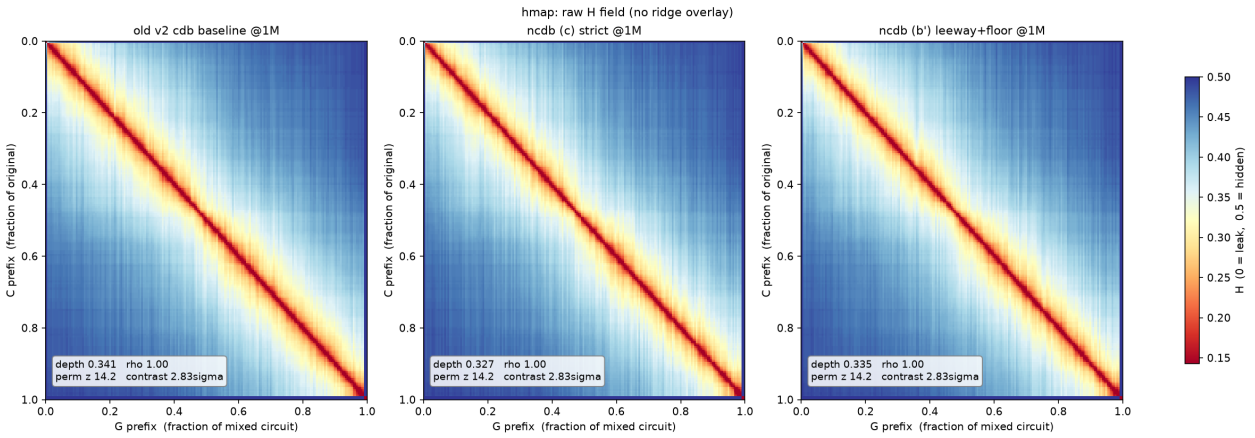


Figure 1: Three-way 1M-move heatmap plates on identical grids: the old v2 store vs. ncdb (c) and (b′).

- **Seed-replicated matrix** (5 stores × 2 seeds, 1M): within-store seed spread 0.003–0.007 equals every between-store gap—no ncdb variant separates from any other; the robust fact is 10/10 ncdb runs below the old store.

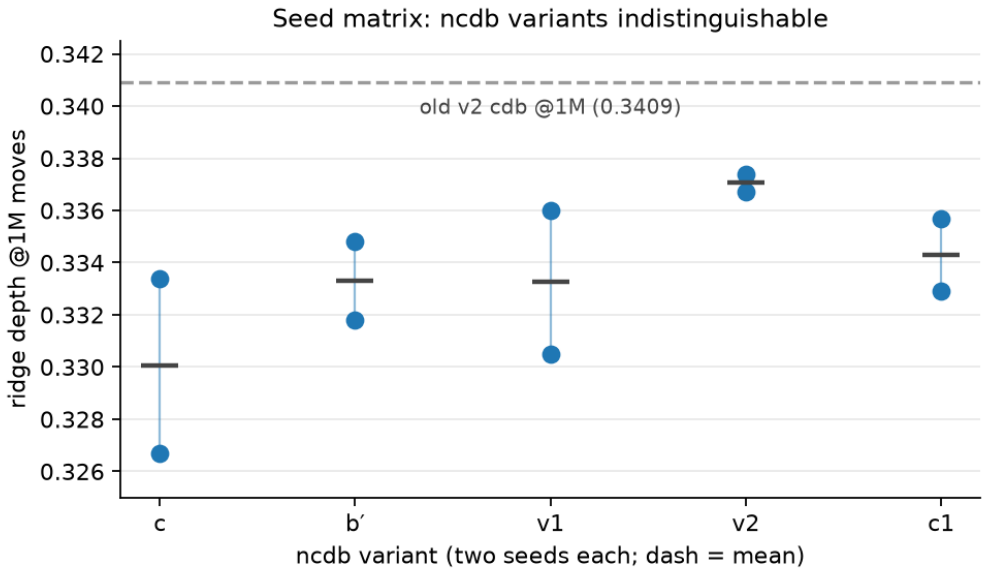


Figure 2: Seed-replicated 1M matrix: each variant's two seeds (dash = mean) against the old store's depth. Within-store spread equals between-store gaps.

- **Expansion-factor sweep** (profile $3, 5, 5, F, F$ on (c)): monotone and strong—$F = 1.5/2.5/3/5 \rightarrow$ depth 0.3243/0.2997/0.2964/0.2814, DB-splices 12.2/24.6/29.5/40.5% of moves (6–14× seed noise); re-confirms "r is the master knob".

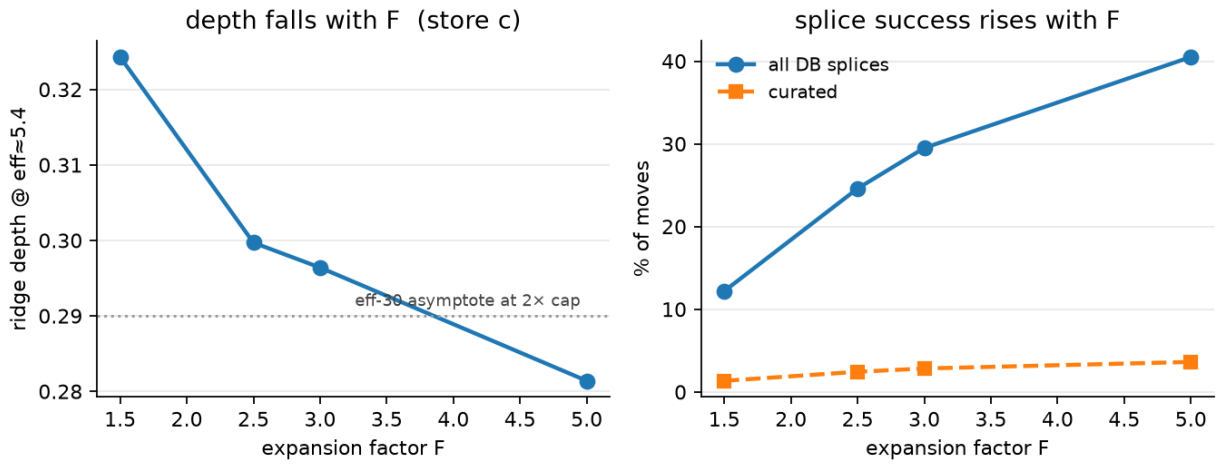


Figure 3: Expansion-factor sweep on store (c): depth falls and splice success rises monotonically with F.

- **Long holds to eff 30** (2× cap, 41.9M moves/arm): (c) and (b′) identical at every point (0.314 → 0.290 vs 0.315 → 0.292), decelerating to an asymptote ≈0.290; splice traffic decays 23% → 10% of moves while curated's *share* grows all run (b′: 30% → 49% of DB splices—self-seeding, later shown causally).

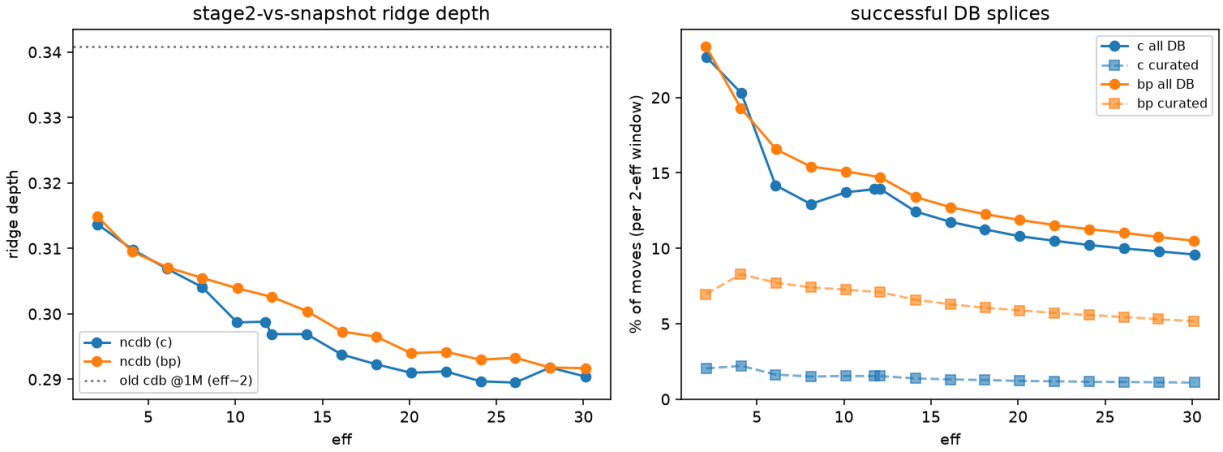


Figure 4: Long holds at the 2× cap: ridge depth (left) and DB-splice fraction (right) vs. eff for (c) and (b′) — the stores never separate; depth asymptotes near 0.290.

- **Ablation (peer runs, random material).** Removing the cdb entirely—or forcing every MIX splice through it at matched splice budget—leaves ridge depth unchanged ($\max \Delta = 0.0035 <$ seed spread). *Affine-ridge depth is pure dilution and is not the axis on which any curated store differentiates*; the follow-on erasability work ties the floor to the input's hull-erasability instead. The cdb's value therefore rests on what these experiments did measure: correctness of served material, certified equal-length traffic for TDP vocabulary, bounded per-key cost, and ∼50× throughput.

6 Why (c) and (b′)

- **(a) is strictly dominated by (c).** Every (a)-pair is in (c) with the identical per-pair guarantee; (c) additionally serves junky keys—the common case in a mixed circuit, and the population the old store punished with self-hits.
- **(b) is dominated by (b′).** Its extra 131M pairs are precisely friends whose small windows compress by 1–2 gates—visibly cancellable junk (entire cohorts, e.g. the single-gate pool's 7-gate friends, exist only through that hazard); no measured benefit offsets it.
- **(c) vs (b′) isolates the one real dial.** Shared key policy and small-window hygiene; they differ only in the graded leeway on spread and large windows—strictest-certified vs maximal-certified-clean, 3.3M vs 63.1M pairs, 1× vs ∼5× curated traffic. The interpolating variants (v1, v2, c1) were built and measured and separate from neither pole: storage without information.
- **Empirically interchangeable, operationally distinct.** On every security metric tried they tie (and the ablation result says the ridge cannot prefer either), so the choice is structural: **(c)** for the purest guarantee at minimal size; **(b′)** when curated-move volume matters (TDP digestion, self-seeding studies, COMP-side traffic). Both frozen stores are validated, filtered, and selectable per run via `FROZEN_CURATED_DIR`.

7 Artifacts

Stores under `frozen_db/`: pools `curated_pools_v3_20260929`; composites `tf_v3_{a,c,b,bp,v1,v2,c1}_*`; frozen `frozen_tf_{c,bp,v1,v2,c1}_*` with filters. Tools (working tree, uncommitted): `from-frozen-identities` `pool_census`, `true_friend_probe` (probe + 7-mode builder + layered oracle), the runtime circuit-key switch with tests, and the run scripts under `runs/ncdb_ab_20261001/`. Open items: the `--tdp` preset stomps `--no-curated*` (flag-pair workaround); store uploads to object storage pending; the identity-function pool ($\delta = 0$) stays excluded by policy.